
DREAM: Dual-Standard Semantic Homogeneity with Dynamic Optimization for Graph Learning with Label Noise

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Abstract

Graph neural networks (GNNs) have been widely used in various graph machine learning scenarios. Existing literature primarily assumes well-annotated training graphs, while the reliability of labels is not guaranteed in real-world scenarios. Recently, efforts have been made to address the problem of graph learning with label noise. However, existing methods often (i) struggle to distinguish between reliable and unreliable nodes, and (ii) overlook the relational information embedded in the graph topology. To tackle this problem, this paper proposes a novel method, Dual-Standard Semantic Homogeneity with Dynamic Optimization (DREAM), for reliable, relation-informed optimization on graphs with label noise. Specifically, we design a relation-informed dynamic optimization framework that iteratively reevaluates the reliability of each labeled node in the graph during the optimization process according to the relation of the target node and other nodes. To measure this relation comprehensively, we propose a dual-standard selection strategy that selects a set of anchor nodes based on both node proximity and graph topology. Subsequently, we compute the semantic homogeneity between the target node and the anchor

nodes, which serves as guidance for optimization. We also provide a rigorous theoretical analysis to justify the design of DREAM. Extensive experiments are performed on six graph datasets across various domains under three types of graph label noise against competing baselines, and the results demonstrate the effectiveness of the proposed DREAM. Our code is available at <https://github.com/YushengZhao/DREAM>.

1. Introduction

Graph neural networks (GNNs) (Kipf & Welling, 2016; Li et al., 2018a; Wu et al., 2019a) have become the primary paradigm for learning representations on non-Euclidean graph-structured data (Ju et al., 2024a; Ma et al., 2025; Zhao et al., 2025b), enabling a plethora of applications in various fields, including recommendation systems (Wu et al., 2019b; Yang et al., 2021; Wang et al., 2024a; Ou et al., 2025), social network analysis (Zhang et al., 2022b; Yin et al., 2022; Jain et al., 2023; Li et al., 2023), spatio-temporal forecasting (Wang et al., 2022; Lan et al., 2022; Ju et al., 2024b; Hu et al., 2025), and protein-protein interactions (Shen et al., 2021; Jha et al., 2022; Réau et al., 2023; Shuvo & Bhattacharya, 2025). Despite their superior performance, most research in this field assumes that graphs are well-annotated, whereas in reality, annotation noise is inevitable and harms model performance (Frénay & Verleysen, 2013; Johnson & Khoshgoftaar, 2022; Wang et al., 2024b).

Recently, the task of graph learning with label noise has received increasing attention, with a number of methods proposed (Dai et al., 2021; 2022; Yin et al., 2023; Zhu et al., 2024; Wu et al., 2025). One line of research focuses on regularizing the loss function through information from the graph structure (Dai et al., 2021; Du et al., 2021; Li et al., 2024c; Wu et al., 2024) or node semantics (Lu et al., 2024; Zhu et al., 2024). Another line of research reweights the samples according to their reliability, typically measured by the loss function (Gui et al., 2021; Wei et al., 2022; Li et al., 2024a; Liao et al., 2025), where a smaller loss indicates reliability of the sample (Yu et al., 2019; Kim et al., 2021).

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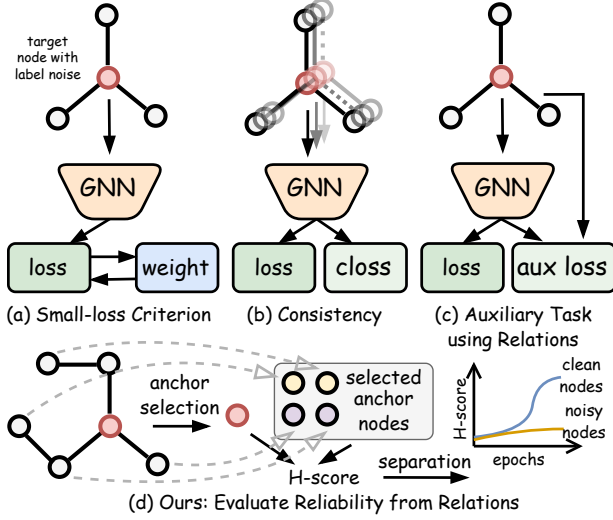


Figure 1. Conventional methods either overlook the relational information in distinguishing noisy labels (e.g., small loss criterion), or resort to regularization for extra supervision signals that are hard to separate clean from noisy nodes (e.g., consistency regularization, auxiliary tasks using edges). By comparison, this paper explicitly evaluates the reliability of nodes from carefully selected relational information in the graph topology.

However, prior research often suffers from two major shortcomings. (i) *Existing works frequently struggle to differentiate between reliable and unreliable nodes within complex graph topology.* When faced with potential label noise, many approaches resort to auxiliary supervision signals, such as contrastive learning (Xue et al., 2022; Zhu et al., 2024) or regularization (Du et al., 2021; Dai et al., 2021), while either falling short in or altogether avoiding the task of distinguishing reliable and unreliable nodes in the complex graph topology. Explicit identification of these nodes can not only mitigate the impact of noisy labels but also prioritize the importance of nodes, benefiting model optimization (Hu et al., 2020a; Zhang et al., 2022a; Yuan et al., 2024). (ii) *Prior works often neglect the rich relational information naturally embedded in the graph topology.* Many works reweight the importance of samples following the small-loss criterion (Liu & Tao, 2015; Gui et al., 2021; Dai et al., 2022; Chen et al., 2024b). However, they often ignore the relational information embedded in the graph topology, which is beneficial to identifying the reliability and informativeness of labeled nodes (Chen et al., 2021; Kim et al., 2023).

To address these limitations, this paper proposes a novel method named Dual-Standard Semantic Homogeneity with Dynamic Optimization (DREAM) for reliable and relation-informed dynamic optimization of graph machine learning models in the presence of label noise. As illustrated in Figure 1, the core idea of DREAM is to leverage the rich relational information within the graph to reevaluate the reliability of each labeled node in the graph iteratively during

optimization. To achieve this goal, we design dual-standard semantic homogeneity that measures the homogeneity between each target node and a set of carefully selected anchor nodes. Specifically, DREAM adopts both proximity-aware and topology-aware anchor selection to identify relevant candidate nodes that form the anchor set for each target node. Subsequently, the homogeneity score between the target node and its anchors is computed based on the similarity of their latent representations obtained during the forward pass. Finally, these homogeneity scores are used as indicators to reweight the importance of each labeled node during optimization. This process is integrated into the model’s supervision and performed iteratively, enabling relation-informed dynamic optimization. We also provide a rigorous theoretical analysis of DREAM, showing that an ϵ -error approximation to the optimal parameters is guaranteed under the proposed training framework. We perform extensive experiments across six graph datasets from different domains under three different types of label noise. The results compared with a number of baselines validate the superiority of the proposed DREAM.

The contributions of this paper are summarized as follows:

- **New Perspective:** We propose to utilize the rich relational information naturally embedded in the graph topology to dynamically re-evaluate the reliability of each labeled node during optimization.
- **Novel Methodology:** We design a novel method, termed dual-standard semantic homogeneity with dynamic optimization (DREAM), which measures the reliability of each labeled node through a set of node-specific anchors selected based on both proximity and graph topology.
- **Theoretical Insights:** We provide a rigorous theoretical analysis, showing that the proposed DREAM achieves an ϵ -error approximation to the optimal model parameters.
- **Extensive Experiments:** We perform extensive experiments across various datasets and different types of label noise against competing baselines, and the results confirm the effectiveness of the proposed DREAM.

2. Related Works

2.1. Graph Neural Networks

Graph neural networks (GNNs) (Kipf & Welling, 2016; Hamilton et al., 2017; Veličković et al., 2017; Xu et al., 2018) have been extensively applied in many graph machine learning tasks (Wu et al., 2020; Corso et al., 2024; Zhao et al., 2025a;c; Wang et al., 2025), facilitating downstream applications in various fields including recommendation (Wu et al., 2019b; Ou et al., 2025), healthcare (Li et al., 2020; Paul et al., 2024), transportation (Wang et al., 2020;

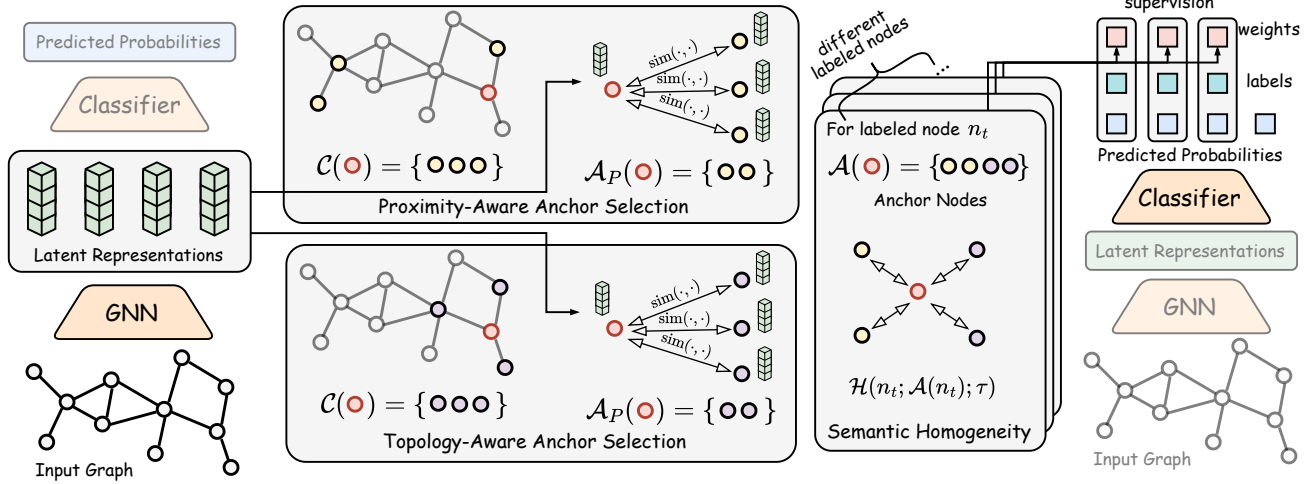


Figure 2. The overall framework of the proposed DREAM. DREAM first selects a set of anchors relative to each target node using proximity-aware anchor selection and topology-aware anchor selection. Then, the semantic homogeneity score is computed according to the relation between the anchors and the target node. Finally, the semantic homogeneity score is used as an indicator of the reliability of each labeled node to reweight the optimization objective dynamically during optimization.

Zhao et al., 2023; Akpaku et al., 2025), and bioinformatics (Wang et al., 2021; Koh et al., 2024; Airas & Zhang, 2025). The graph neural networks obtain the node embeddings by propagating the representations from adjacent nodes iteratively (Xu et al., 2018). Subsequently, the obtained features are used in downstream tasks like classification (Kipf & Welling, 2016) or regression (Jia & Benson, 2020; Zhang et al., 2024). However, most of the models are trained on well-annotated datasets (Hu et al., 2020b; Morris et al., 2020; Dwivedi et al., 2023), whereas in practice, label noise often exists and hurts the optimization of GNNs (Wang et al., 2024b; Zhu et al., 2024; Han et al., 2025). Therefore, this paper studies graph learning with label noise and proposes a relation-informed dynamic optimization framework to train GNNs robustly against noisy labels.

2.2. Learning with Label Noise

Label noise is a widespread problem in deep learning, where part of the training data is incorrectly labeled (Frénay & Verleysen, 2013; Patrini et al., 2017; Lukasik et al., 2020). In practice, many datasets are manually labeled, leading to potential misclassifications of instances (Ekambaram et al., 2017; Yun et al., 2021). This would hurt the optimization of deep learning models. A number of methods have been proposed that focus on Euclidean data like images or texts (Karim et al., 2022; Penso et al., 2024; Jiang et al., 2024). These works often tackle the problem by loss correction (Patrini et al., 2017; Arazo et al., 2019; Toner & Storkey, 2023) or relabeling them for training (Zheng et al., 2021; Huang et al., 2022; Park et al., 2023; Li et al., 2024b).

Nevertheless, the label noise problem in graph learning is

more complex due to the non-Euclidean structure. The label noise on graphs may be uniformly distributed or structure-dependent (Wang et al., 2024b). To address the problem, one line of research regularizes the loss objective with additional supervision signals from self-consistency (Li et al., 2024c; Lu et al., 2024; Yin et al., 2024) or the graph structure (Dai et al., 2021; Wu et al., 2024; Cheng et al., 2024). However, they often struggle to differentiate the noisy labels, hurting the performance when the noise increases. Another line of research reweights the nodes based on certain measurements like the loss function (Dai et al., 2022; Yang et al., 2023; Chen et al., 2024a). However, they often neglect the rich relational information in the graph topology. While there are efforts in label correction, they often rely on assumptions about the graph structure or patterns of mislabeling (Li et al., 2022; Yin et al., 2023; Ju et al., 2026). In this paper, we propose a novel method named DREAM that utilizes the relational information in the graph topology to dynamically reevaluate the reliability of labeled nodes during optimization for robust graph learning with label noise.

3. Methodology

3.1. Problem Definition

We denote a graph as $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, \mathbf{X} \rangle$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, and $\mathbf{X}$ is the input features of nodes. We assume $\mathbf{X} \in \mathbb{R}^{N \times d_{in}}$, where $N = |\mathcal{V}|$ and d_{in} is the dimension of the input node feature. To obtain the node embeddings, a graph neural network f_θ is used, and the learned representations of each node are denoted as $f_\theta(\mathcal{G}) = \mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N]^\top \in \mathbb{R}^{N \times d}$, where d is the hidden dimension. Subsequently, a classifier is g_ϕ used to

obtain the class probabilities p_i of each node. We denote the number of classes as C , the set of nodes that have labels as $\mathcal{S}$, and the noisy labels as $\mathcal{Y} = \{y_i \mid i \in \mathcal{S}\}$. The goal is to optimize the model $\mathcal{M} = f_\theta \circ g_\phi$ with the noisy labels $\mathcal{Y}$ to achieve better performance on the test nodes.

3.2. Framework Overview

We now present the proposed Dual-Standard Semantic Homogeneity with Dynamic Optimization (DREAM), as illustrated in Figure 2. To measure the reliability of each labeled node (the target node, denoted as n_t), DREAM first selects a set of relevant nodes (the anchors, denoted as $\mathcal{A}(n_t) \subset \mathcal{V}$) for each target node n_t . Specifically, we design proximity-aware and topology-aware anchor selection mechanisms to construct the anchor set $\mathcal{A}(n_t)$. Subsequently, we compute a semantic homogeneity score based on the similarity between the target node and its anchors. This score serves as an estimate of the labeled node’s reliability and is used to reweight the optimization objective. During the optimization process, we iteratively reevaluate the reliability of each labeled node using the dual-standard semantic homogeneity, enabling robust dynamic optimization.

3.3. Proximity-Aware Anchor Selection

We first select the anchor nodes according to the semantic proximity of nodes. For a target node n_t that is correctly labeled as y_t , we expect it to have many semantically proximal nodes that are also labeled as y_t , and vice versa. By examining the relationship of the target node and its selected proximal nodes, we can assess the reliability of the target node. Therefore, we propose proximity-aware anchor selection to select these nodes as anchors.

Specifically, given target node n_t labeled as y_t , we first construct the set of candidate nodes that share the label of n_t . Formally, we have the candidate set for n_t as:

$$\mathcal{C}_P(n_t) = \{n_i \mid i \in \mathcal{S} \wedge y_i = y_t\}. \quad (1)$$

We then select the semantically proximal nodes from $\mathcal{C}_P(n_t)$ as proximity-aware anchors. To achieve this, we utilize the learned representations of nodes obtained during the model’s forward pass, *i.e.*, $z_i, i \in \mathcal{C}_P(n_t)$, and a similarity measurement $\text{sim}(\cdot, \cdot)$. Formally, we have:

$$\mathcal{A}_P(n_t) = \underset{\mathcal{I} \subseteq \mathcal{C}_P(n_t), |\mathcal{I}|=k_P}{\text{argmax}} \sum_{i \in \mathcal{I}} \text{sim}(z_t, z_i), \quad (2)$$

where k_P is the number of proximity-aware anchors. Conceivably, for n_t correctly labeled as y_t , the set of $\mathcal{C}_P(n_t)$ contains a number of nodes similar to n_t , as most of them fall into the correct category of y_t . Conversely, when n_t is incorrectly labeled, only a few nodes in $\mathcal{C}_P(n_t)$ fall into the correct category $\hat{y}_t$ of n_t . This observation serves as guidance for evaluating the reliability of the labeled nodes.

3.4. Topology-Aware Anchor Selection

In addition to semantic proximity, the graph topology also plays a significant role. For a target node n_t labeled as y_t , the category of y_t may contain several sub-categories scattered across the graph. Nodes closer in the topology are more likely to belong to the same sub-category. The goal of topology-aware anchor selection is to identify anchors that potentially belong to the same sub-category as a correctly labeled n_t . By comparing the anchors with the target node, we can distinguish reliable nodes from unreliable ones.

Concretely, given two nodes n_i and n_j , we measure their topological proximity using the geodesic distance (Bondy & Murty, 2008), denoted as $d_G(n_i, n_j)$. We then construct the candidate set for n_t according to the topological proximity as follows:

$$\mathcal{C}_T(n_t) = \{n_i \mid d_G(n_i, n_t) \leq d_{max}\}, \quad (3)$$

where d_{max} is the upper bound of geodesic distance from any node to n_t , and we treat it as a hyperparameter.

With the topology-aware candidate set for n_t , we then select the topology-aware anchors similar to Eq. 2 as follows:

$$\mathcal{A}_T(n_t) = \underset{\mathcal{I} \subseteq \mathcal{C}_T(n_t), |\mathcal{I}|=k_T}{\text{argmax}} \sum_{i \in \mathcal{I}} \text{sim}(z_t, z_i), \quad (4)$$

where k_P is the number of proximity-aware anchors. When the size of the candidate set is smaller than k_P (*e.g.*, $|\mathcal{C}_P(n_t)| < k_P$), DREAM uses all available candidates as a fallback, *i.e.*, $\mathcal{A}_P(n_t) = \mathcal{C}_P(n_t)$.

3.5. Relation-Informed Dynamic Optimization

After the proximity-aware and topology-aware anchors have been selected, we can now leverage the semantic homogeneity of the target node n_t and the anchors to determine the reliability of the label y_t . A correctly labeled node will have many semantically similar nodes at both the global categorical level (proximity-aware anchors) and local sub-categorical level (topology-aware anchors). Conversely, for an incorrectly labeled node, the anchor selection processes will include many less similar nodes, as the label y_t does not reflect the node’s true semantics.

Specifically, we first construct the set of anchor nodes relative to n_t as follows:

$$\mathcal{A}(n_t) = \mathcal{A}_P(n_t) \cup \mathcal{A}_T(n_t). \quad (5)$$

Subsequently, we measure the semantic homogeneity of node n_t and the set of anchors according to the relations of the target node and each anchor, *i.e.*,

$$\mathcal{H}(n_t; \mathcal{A}(n_t); \tau) = \left(\frac{1}{|\mathcal{A}(n_t)|} \sum_{i \in \mathcal{A}(n_t)} \text{sim}(z_t, z_i) \right)^{1/\tau}, \quad (6)$$

Algorithm 1 Optimization Algorithm of DREAM

Requires: a $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, \mathbf{X} \rangle$, the set of nodes with labels $\mathcal{S}$, and noisy labels $\mathcal{Y} = \{y_i \mid i \in \mathcal{S}\}$

Ensures: the optimized model $\mathcal{M} = f_\theta \circ g_\phi$.

- 1: Initialize the parameters of model $\mathcal{M}$;
- 2: Compute the candidate sets $\mathcal{C}_P(n_t)$ and $\mathcal{C}_T(n_t)$ for $t \in \mathcal{S}$ according to Eq. 1 and Eq. 3;
- 3: **repeat**
- 4: Compute the latent representations of each node, *i.e.*, $\mathbf{z}_i, i \in \mathcal{V}$ using the encoder f_θ ;
- 5: Select the proximity-aware anchors, *i.e.*, $\mathcal{A}_P(n_t)$, from $\mathcal{C}_P(n_t)$ for $t \in \mathcal{S}$ according to Eq. 2;
- 6: Select the topology-aware anchors, *i.e.*, $\mathcal{A}_T(n_t)$, from $\mathcal{C}_T(n_t)$ for $t \in \mathcal{S}$ according to Eq. 4;
- 7: Obtain the set of anchors using Eq. 5;
- 8: Compute the semantic homogeneity of each labeled node, *i.e.*, $\mathcal{H}(n_t; \mathcal{A}(n_t); \tau)$, using Eq. 6;
- 9: Compute the class probability distributions of each labeled nodes, *i.e.*, $\mathbf{p}_i, i \in \mathcal{S}$, using g_ϕ ;
- 10: Compute the loss function in Eq. 7.
- 11: Back-propagate through the model $\mathcal{M}$ and update the model's parameters.
- 12: **until** convergence

where τ is the temperature for scaling. Finally, we use the semantic homogeneity as a measurement of reliability to reweight the vanilla cross-entropy objective:

$$\mathcal{L} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathcal{H}(n_i; \mathcal{A}(n_i); \tau) \cdot \text{CE}(\mathbf{p}_i, y_i), \quad (7)$$

where $\text{CE}(\cdot, \cdot)$ denotes the cross-entropy loss.

Moreover, since anchor selection relies on latent representations encoded by f_θ , we design a dynamic optimization process that reevaluates the reliability of each labeled node during each epoch. In this way, re-evaluating reliability leads to improved supervision signals (as defined in Eq. 7); better supervision leads to more accurate node representations, which in turn enables better anchor selection—ultimately enhancing the model's optimization process. The overall algorithm is presented in Algorithm 1

4. Theoretical Analysis

In this section, we aim to provide a theoretical understanding of the intuition behind our proposed DREAM. Before going into the details, for any node $i \in \mathcal{S}$, we denote its ground-truth label as $\hat{y}_i$ and represent the corresponding input feature vector as $\mathbf{x}_i \in \mathbb{R}^{1 \times d_{in}}$. Moreover, we assume that each input-label pair $\hat{\xi}_i \triangleq (\mathbf{x}_i, \hat{y}_i)$ is independently drawn from an underlying distribution $\mathcal{P}$.

Generally speaking, the primary goal of our DREAM is

to learn a probability function $\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i)$ such that we can achieve accurate node classification, where $\mathbf{w} \triangleq (\theta, \phi)$ denotes all model parameters. To this aim, we often consider the following stochastic optimization problem:

$$\min_{\mathbf{w}} F(\mathbf{w}) \triangleq \mathbb{E}_{\hat{\xi}_i = (\mathbf{x}_i, \hat{y}_i) \sim \mathcal{P}} [\text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), \hat{y}_i)] \quad (8)$$

However, in many real-world scenarios, due to factors such as human annotation error and improper data collection processes, the observed label y_i is often corrupted causing the obtained input-label pair $\xi_i \triangleq (\mathbf{x}_i, y_i)$ to follow a different distribution, denoted as $\mathcal{Q}$.

Luckily, according to the change-of-measure rules (Liu & Tao, 2015) or the Radon–Nikodym theorem (Durrett, 2019), we can rewrite the objective $F(\mathbf{w})$ in Eq. 8 as an expectation over the distribution $\mathcal{Q}$, that is to say,

$$\begin{aligned} F(\mathbf{w}) &\triangleq \mathbb{E}_{\hat{\xi}_i = (\mathbf{x}_i, \hat{y}_i) \sim \mathcal{P}} [\text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), \hat{y}_i)] \\ &= \mathbb{E}_{\xi_i = (\mathbf{x}_i, y_i) \sim \mathcal{Q}} \left[\frac{\Pr((\mathbf{x}_i, \hat{y}_i) | \mathcal{P})}{\Pr((\mathbf{x}_i, y_i) | \mathcal{Q})} \cdot \text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), y_i) \right] \\ &= \mathbb{E}_{\xi_i = (\mathbf{x}_i, y_i) \sim \mathcal{Q}} [\alpha(\mathbf{x}_i, y_i, \hat{y}_i) \cdot \text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), y_i)], \end{aligned} \quad (9)$$

where $\alpha(\mathbf{x}_i, y_i, \hat{y}_i) \triangleq \frac{\Pr((\mathbf{x}_i, \hat{y}_i) | \mathcal{P})}{\Pr((\mathbf{x}_i, y_i) | \mathcal{Q})}$ is the ratio of distribution drift and the symbol $\Pr(\cdot | \mathcal{P})$ or $\Pr(\cdot | \mathcal{Q})$ represents the probability under the distribution $\mathcal{P}$ or $\mathcal{Q}$.

Inspired by the results of Eq. 9, our proposed DREAM algorithm adopts a weighted cross-entropy loss (Eq. 7) to learn the model parameter $\mathbf{w}$. More importantly, for each node $i \in \mathcal{S}$, DREAM introduces a novel semantic homogeneity score $\mathcal{H}(n_i; \mathcal{A}(n_i); \tau)$ to approximate the unknown ratio of distribution drift $\alpha(\mathbf{x}_i, \hat{y}_i, y_i)$. In sharp contrast to previous studies, this homogeneity score fully exploits the rich relational information naturally embedded in the graph topology. Building on these findings, we next investigate the convergence of our DREAM algorithm.

Before that, we first make some standard assumptions about the objective function $F(\mathbf{w})$, that is,

Assumption 4.1 (Smoothness). The objective function $F(\mathbf{w})$ is L -smooth, that is, $F(\mathbf{w})$ is differentiable and there exists a constant $L > 0$ such that

$$\|\nabla F(\mathbf{w}) - \nabla F(\mathbf{u})\| \leq L \|\mathbf{w} - \mathbf{u}\|_2,$$

where the symbol $\|\cdot\|_2$ denotes l_2 norm.

Assumption 4.2 (Polyak–Łojasiewicz Condition). There exists a constant $\mu > 0$ such that

$$2\mu(F(\mathbf{w}) - F(\mathbf{w}^*)) \leq \|\nabla F(\mathbf{w})\|_2^2,$$

where $\mathbf{w}^* \triangleq \arg \min_{\mathbf{w}} F(\mathbf{w})$.

Remark 4.3. It is important to highlight that the Polyak-Łojasiewicz condition has been theoretically and empirically verified during the training of deep neural networks (Li et al., 2018b; Wang & Van Hoof, 2022; Yue et al., 2023).

Assumption 4.4 (Boundedness). For any node $i \in \mathcal{S}$, the gradient $\nabla_{\mathbf{w}} \text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), y_i)$ is bounded, that is, $\|\nabla_{\mathbf{w}} \text{CE}(\mathbf{p}_i(\mathbf{w}; \mathbf{x}_i), y_i)\|_2 \leq G$.

Furthermore, we suppose, at every outer iteration k of Algorithm 1, the model parameters $\mathbf{w}$ are updated using standard gradient descent, i.e., $\mathbf{w}_{k+1} \triangleq \mathbf{w}_k - \eta \nabla \mathcal{L}(\mathbf{w}_k)$, where η denotes the step size and $\mathcal{L}(\mathbf{w})$ represents the loss function utilized by our DREAM method (See Eq. 7). With all these preparations, we can have the following theorem:

Theorem 4.5 (Proof is deferred to Appendix). *Under Assumption 4.1-4.4, if the semantic homogeneity $\mathcal{H}(n_i; \mathcal{A}(n_i); \tau)$ of any node $i \in \mathcal{S}$ is a (β, ϵ) -approximation to the $\alpha(\mathbf{x}_i, \hat{y}_i, y_i)$, i.e., $|\mathcal{H}(n_i; \mathcal{A}(n_i); \tau) - \beta \cdot \alpha(\mathbf{x}_i, \hat{y}_i, y_i)| \leq \epsilon$, the step size $\eta \triangleq \mathcal{O}(\frac{1}{\beta L})$ and the size of labeled nodes $|\mathcal{S}| \triangleq \mathcal{O}(\frac{1}{\epsilon})$, after k iterations, the final model parameters $\mathbf{w}_{k+1}$ yielded by our proposed DREAM satisfies that:*

$$\mathbb{E}(F(\mathbf{w}_{k+1}) - F(\mathbf{w}^*)) \leq \mathcal{O}\left(\left(1 - \frac{\mu}{3L}\right)^k + \frac{\beta^2 G^2}{\mu} \epsilon\right),$$

where $\mathbf{w}^*$ is the optimal parameters setup.

Remark 4.6. Theorem 4.5 implies that when the relational information embedded in the graph is sufficiently rich to estimate the unknown ratio of distribution drift $\alpha(\mathbf{x}_i, \hat{y}_i, y_i)$, if we set $\eta \triangleq \mathcal{O}(\frac{1}{\beta L})$ and $|\mathcal{S}| \triangleq \mathcal{O}(\frac{1}{\epsilon})$, after $\mathcal{O}(\frac{L}{\mu} \log(\frac{1}{\epsilon}))$ iterations, our proposed DREAM method can achieve an ϵ -error approximation to the optimal model parameters $\mathbf{w}^*$.

5. Experiments

5.1. Experimental Setup

Datasets. Our evaluation is conducted on six widely used graph benchmark datasets across various domains, including Cora (McCallum et al., 2000), CiteSeer (Giles et al., 1998), PubMed (Sen et al., 2008), DBLP (Pan et al., 2016), A-Photo (Shchur et al., 2018), and Flickr (Yang et al., 2020). Both homophilic and heterophilic graphs are included. More details about the datasets can be found in Appendix B.

Compared Baselines. The proposed method is compared against a comprehensive suite of state-of-the-art baselines, categorized as follows. (1) *Vanilla GNN*, including GCN (Kipf & Welling, 2016). (2) *Learning with Label Noise (LLN) methods*, including S-model (Goldberger & Ben-Reuven, 2017), Coteaching (Han et al., 2018), JoCoR (Wei et al., 2020), APL (Ma et al., 2020), SCE (Wang et al., 2019), Forward (Patrini et al., 2017), and Backward (Patrini et al., 2017). (3) *Graph Neural Networks under Label Noise*

(GLN) methods, including NRGNN (Dai et al., 2021), RT-GNN (Qian et al., 2023), CP (Zhang et al., 2020), CLNode (Wei et al., 2023), PIGNN (Du et al., 2021), DGNN (NT et al., 2019), RNCGLN (Zhu et al., 2024), UnionNET (Li et al., 2021), CGNN (Yuan et al., 2023), and CR-GNN (Li et al., 2024c). More details are listed in Appendix C.

Evaluation Details. In this paper, we generally follow the evaluation setting of NoisyGL (Wang et al., 2024b). The labels in the training and validation sets are subject to noise, while the test set remains clean for proper evaluation. In addition to the uniform label noise, we also adopt pair noise and asymmetric noise in NoisyGL (Wang et al., 2024b). We adopt the noise rate of 30% for all noise types by default and report the average classification accuracies over ten runs.

Implementation Details. We adopt a two-layer GCN (Kipf & Welling, 2016) as the graph learning model $\mathcal{M}$. f_θ and g_ϕ are both implemented with one graph convolution layer, and the hidden dimension d is fixed to 64. We also adopt contrastive learning (You et al., 2020) following prior works (Yuan et al., 2023; Li et al., 2024c), as described in Appendix D. For the similarity measurement, i.e., $\text{sim}(\cdot, \cdot)$, we use cosine similarity rescaled to the interval of $[0, 1]$. By default, we set the hyperparameters as follows. In proximity-aware anchor selection, we set the number of anchors k_P to 15. In topology-aware anchor selection, we set the number of anchors k_T to 10. The upper bound of geodesic distance d_{max} is set to 4. During the computation of semantic homogeneity, we set the temperature τ to 0.04. For optimization, we use Adam optimizer (Kingma, 2014) with a learning rate of 1×10^{-2} and train the model for 500 epochs on an NVIDIA H800 GPU.

5.2. Main Results

We report the average classification accuracy of the proposed DREAM compared to baseline methods over ten runs, and the results are shown in Table 1. According to the results, we have the following observations. Firstly, the proposed DREAM achieves a consistent improvement in terms of accuracy over existing baselines on all six datasets under different types of label noise. Specifically, DREAM achieves 14.9% relative improvement compared to the second best on the CiteSeer dataset with uniform label noise. The significant improvements demonstrate the robustness of DREAM against label noise. Secondly, learning with label noise (LLN) methods designed for Euclidean data (e.g., JoCoR, APL, SCE) generally perform similar to or worse than vanilla GCN. One possible explanation is that they do not consider the topological information in the graphs under label noise. Lastly, for graph neural networks under label noise (GLN) methods, such as NRGNN, RT-GNN, PIGNN, while some exhibit mild improvement over vanilla GCN, their overall performance remains unsatis-

Table 1. The classification accuracy on six benchmark datasets. The best results are shown in **bold** and second-best underline.

Dataset	Cora			CiteSeer			PubMed			DBLP			A-Photo			Flickr		
Noise Type	pair	uniform	asym.	pair	uniform	asym.	pair	uniform	asym.	pair	uniform	asym.	pair	uniform	asym.	pair	uniform	asym.
GCN	65.36	71.06	69.72	53.66	55.05	56.47	62.91	66.53	66.53	62.56	69.66	66.60	79.26	84.86	82.23	45.86	<u>49.77</u>	<u>49.54</u>
S-model	65.59	71.05	70.04	53.41	54.68	56.28	63.16	66.52	66.89	62.60	69.68	66.37	80.06	85.32	84.22	45.47	49.42	49.27
Coteaching	56.87	61.22	62.09	48.34	46.75	47.47	60.45	64.38	61.76	56.98	64.60	61.99	74.56	82.78	77.84	41.55	43.53	44.19
JoCoR	64.52	71.16	70.84	56.30	57.14	59.05	55.88	59.44	57.27	62.39	68.60	66.33	68.48	75.22	79.51	41.20	43.61	44.17
APL	65.97	72.37	70.67	53.48	54.61	56.02	64.28	67.56	67.27	64.25	69.37	66.81	82.39	<u>86.17</u>	<u>85.27</u>	40.71	45.64	43.44
SCE	66.97	71.15	70.17	54.61	57.00	57.19	63.78	67.38	69.02	63.86	70.18	67.86	81.57	85.85	82.83	23.36	22.69	24.32
Forward	65.96	72.47	70.01	54.57	56.05	56.94	62.68	65.62	65.97	63.59	70.27	67.03	67.29	65.13	63.60	22.98	21.49	19.88
Backward	65.64	72.06	70.05	54.62	55.81	56.93	62.35	65.61	65.84	63.59	70.26	66.72	51.90	55.97	54.90	25.40	23.93	22.74
NRGNN	<u>69.73</u>	<u>74.86</u>	<u>74.48</u>	<u>59.28</u>	<u>61.87</u>	63.33	57.22	62.37	62.38	<u>69.82</u>	<u>76.21</u>	<u>73.83</u>	71.92	78.30	72.38	38.79	42.76	44.52
RTGNN	63.44	66.33	66.84	48.39	51.38	52.57	59.04	68.30	66.72	59.26	66.89	65.59	70.33	82.08	83.68	36.50	43.72	40.85
CP	66.12	71.76	71.09	54.15	56.72	57.71	62.67	68.11	68.17	65.80	70.49	68.30	77.45	83.52	82.02	40.74	43.13	42.54
CLNode	64.55	68.24	67.61	53.19	55.48	55.24	62.25	64.56	66.20	62.59	65.31	62.45	79.12	85.46	82.57	45.06	48.06	47.45
PIGNN	65.04	70.43	69.03	55.97	60.71	59.43	<u>66.75</u>	<u>69.02</u>	70.50	69.31	71.98	70.76	79.43	84.90	83.75	44.61	49.38	49.06
DGNN	58.49	56.15	57.16	46.76	48.99	49.39	58.36	63.29	63.42	61.99	63.47	61.45	51.09	54.28	56.66	15.64	19.24	15.80
RNCGLN	68.48	73.29	74.35	54.16	58.56	<u>66.12</u>	63.70	68.50	<u>70.90</u>	64.35	63.65	65.54	64.79	72.80	74.37	22.95	23.99	24.38
UnionNET	66.29	72.83	71.02	58.15	58.87	60.85	62.77	65.93	65.60	62.92	70.33	66.97	30.89	32.29	27.53	20.75	21.19	22.41
CGNN	63.27	69.57	69.01	51.38	52.31	51.09	56.01	57.73	59.76	58.06	63.26	58.70	45.70	47.26	49.43	11.33	11.71	11.73
CR-GNN	65.37	72.03	70.40	52.45	55.87	53.92	62.91	66.42	69.48	63.79	70.23	66.94	37.53	35.24	31.77	29.38	31.09	30.19
DREAM	74.84	80.40	79.40	66.49	71.06	71.20	69.06	71.75	72.45	76.57	79.06	79.34	83.20	89.19	87.86	48.39	53.84	53.55

Table 2. Ablation studies of the proposed DREAM.

Dataset	Cora			CiteSeer			PubMed		
Noise	pair	uniform	asym.	pair	uniform	asym.	pair	uniform	asym.
GCN	65.36	71.06	69.72	53.66	55.05	56.47	62.91	66.53	66.53
V1	74.38	79.60	78.89	65.42	69.69	70.07	68.77	71.48	72.24
V2	70.56	77.04	75.56	59.90	62.11	62.77	67.38	69.88	70.39
V3	66.07	71.78	70.40	54.13	56.59	56.98	62.87	66.35	66.93
V4	70.85	75.81	75.56	61.64	65.96	66.86	67.93	70.48	72.06
V5	68.88	74.77	74.56	60.85	65.16	66.11	68.29	70.62	72.07
DREAM	74.84	80.40	79.40	66.49	71.06	71.20	69.06	71.75	72.45

factory—particularly on heterophilic graphs (e.g., Flickr). These methods often struggle to distinguish reliable from unreliable nodes and fail to effectively utilize the relational information embedded in the graph topology.

5.3. Ablation Studies

We then investigate how the different components of DREAM affect overall performance and present the results in Table 2. Specifically, we design five variants of DREAM. V1 removes the topology-aware anchors (i.e., $\mathcal{A}_T(n_t)$) from the DREAM framework. V2 removes the proximity-aware anchors (i.e., $\mathcal{A}_P(n_t)$). V3 disables temperature scaling (i.e., $\tau = 1$) when computing semantic homogeneity scores. V4 removes the candidate sets (i.e., $\mathcal{C}_P(n_t)$ and $\mathcal{C}_T(n_t)$) and selects anchors from all the nodes $\mathcal{V}$. V5 selects from the union of both candidate sets (i.e., $\mathcal{C}_P(n_t) \cup \mathcal{C}_T(n_t)$). As can be seen from the results, each component of DREAM contributes to overall performance, and removing or replacing any of them with alternatives degrades performance. Temperature scaling is an important technique in DREAM, as is demonstrated by low performance (V3) when it is disabled. One explanation is that scaling enlarges the gap between clean (reliable) and noisy (unreliable) nodes, making training more effective. Additionally, selecting anchors from candidates separately (proximity-aware and topology-aware) is also important, as suggested by the low accuracy

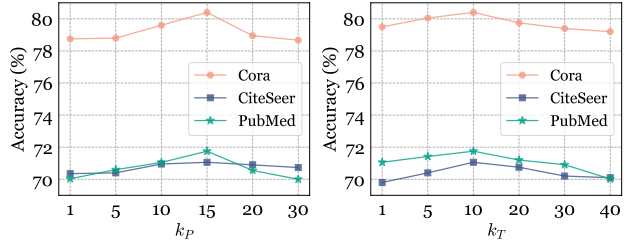


Figure 3. Hyperparameter analysis of DREAM on three datasets (i.e., Cora, CiteSeer, and PubMed). We show the accuracy under different numbers of proximity-aware anchors (i.e., k_P , left) and topology-aware anchors (i.e., k_T , right).

in V4 and V5. The candidate sets provide proximity and topology awareness, filtering out irrelevant nodes less meaningful for the target nodes, while separate selection ensures balanced sampling from both sets of candidates.

5.4. Hyperparameter Analysis

We also show how the proposed DREAM performs under different hyperparameters. The results on three datasets under 30% uniform noise are presented in Figure 3.

Numbers of Proximity-Aware Anchors. The prediction accuracy of DREAM under different numbers of proximity-aware anchors (i.e., k_P) is shown in Figure 3 (left). As can be seen from the results, DREAM achieves best accuracy when k_P is set to 15. When the number of proximity-aware anchors is too small, the model may not have enough proximity awareness for deciding the reliability of the target nodes. Conversely, when k_P is too large, the selection process may include more noise from additional anchors.

Numbers of Topology-aware Anchors. We also present the effect of the number of topology-aware anchors (i.e., k_T) in Figure 3 (right). As can be seen from the figure, our

Table 3. Average training time (in seconds) over 10 runs.

Dataset	Cora	CiteSeer	PubMed	DBLP	A-Photo	Flickr
NRGNN	4.52	4.36	72.23	27.03	41.63	40.15
RTGNN	7.95	16.13	73.62	65.46	98.66	73.87
UnionNET	26.78	13.54	9.05	10.75	37.36	66.11
PIGNN	3.84	11.83	52.79	6.74	22.17	34.06
DREAM	3.66	4.17	19.40	12.25	6.84	12.45

method achieves the best performance when k_T is set to 10. Similar to hyperparameter k_P , the change of k_T affects the model’s topology awareness in the computation of semantic homogeneity, in addition to the balance between proximity awareness and topology awareness. A moderate value of 10 topology-aware anchors leads to the best result.

5.5. Further Analysis

Model Efficiency. We also evaluate the efficiency of the proposed DREAM in comparison with various baselines, as shown in Table 3, which reports the training time in seconds over ten runs. As can be seen from the data, the proposed method is relatively efficient compared to baselines, exhibiting similar or smaller training time. Additionally, the proposed DREAM is significantly faster than baselines on dense graphs—specifically, A-Photo (average degree: 31.13) and Flickr (average degree: 63.30). We attribute this to our design of candidate sets (Eq. 1 and Eq. 3), which are precomputed before training and filter out less relevant nodes from the selection of anchors.

Evolution of Semantic Homogeneity Score. We then show that DREAM can differentiate nodes with clean labels from nodes with noisy labels dynamically during optimization. Specifically, we compute the average semantic homogeneity score in Eq. 6 of nodes with clean labels and nodes with noisy labels. We present the changes in average scores of both groups of nodes during model optimization on the Cora and CiteSeer datasets under 30% uniform label noise in Figure 4. As shown in the figure, the proposed DREAM effectively separates clean nodes from noisy ones, and this separation is achieved dynamically throughout the optimization process. During the initial epochs of training, the average semantic homogeneity scores of clean and noisy nodes are similar. As the optimization progresses, the average scores quickly diverge, and the gap generally widens over the course of training. As the homogeneity scores are re-computed in each epoch, the weights measuring reliability dynamically change during optimization, forming a positive feedback loop: better measurement of reliability leads to improved supervision signals, and the resulting enhancement of the model enables better selection of anchors, which in turn further improves the measurement of reliability through the computed semantic homogeneity scores.

Performance Under Different Noise Levels. We also show that DREAM performs consistently better than baseline

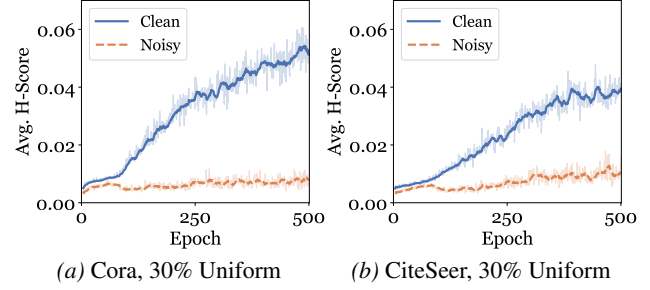


Figure 4. Evolution of semantic homogeneity scores of clean and noisy nodes on Cora and CiteSeer under 30% uniform label noise. DREAM successfully separates clean from noisy nodes.

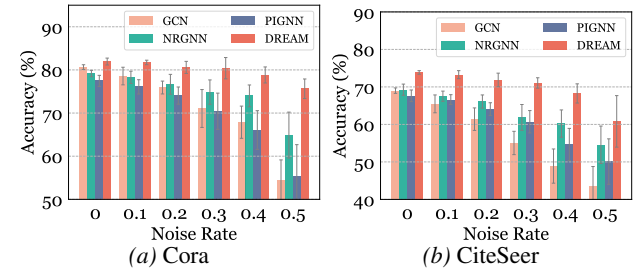


Figure 5. Performance under different levels of uniform label noise on (a) Cora and (b) CiteSeer. The accuracy of the proposed DREAM degrades significantly slower than baseline methods as the noise rate increases from 0 to 0.5.

methods under different noise levels. Specifically, we train our model and baselines under different noise rates ranging from 0 to 0.5, and compare the classification accuracy on the clean test set. The results on the Cora and CiteSeer datasets are presented in Figure 5, with additional results under other settings deferred to Appendix E.1, where similar results can be observed. From the results, we can see that DREAM consistently outperforms the baselines under different noise rates on both datasets. Additionally, as the ratio of noisy labels increases, our method still achieves strong accuracy, and the performance gap between the proposed DREAM and the baselines generally widens. This suggests that our method is more robust against large noise rates and serves as a more practical solution under higher levels of label noise.

Visualization of Learned Representations. We also provide visualization of learned node representations using t-SNE (Maaten & Hinton, 2008). Specifically, we compare the node representations generated by GCN and the proposed DREAM on the Cora dataset under 40% uniform label noise, and the results are presented in Figure 6. As can be seen from the figure, DREAM generates denser and more clustered node embeddings, e.g., the blue dots in Figure 6 (b) compared to (a). This shows that while the node embeddings learned by GCN are hampered by noisy labels, the proposed DREAM reduces the negative effect of label noise through the estimated reliability from anchor nodes.

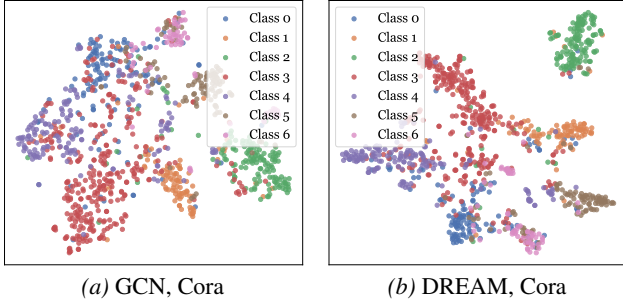


Figure 6. Learned representations under 40% uniform label noise on Cora. The proposed DREAM yields more condensed clusters and better separates different classes compared to GCN.

6. Conclusion

This paper investigates the important problem of graph learning under label noise and proposes a novel framework named Dual-Standard Semantic Homogeneity with Dynamic Optimization (DREAM) to enable reliable relation-informed dynamic optimization of graph learning models. In particular, we measure the reliability of each labeled node based on its relationship with a set of anchors carefully selected according to both semantic proximity and graph topology. Semantic homogeneity scores are computed to quantitatively measure the reliability and are used to reweight the optimization objective. We provide a rigorous theoretical analysis as well as extensive empirical experiments to demonstrate the effectiveness of DREAM.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Proof of Theorem 4.5

In this section, we provide a proof of Theorem 4.5. At first, from Eq.(7), we have that

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathcal{H}(n_i; \mathcal{A}(n_i); \tau) \nabla \text{CE}(\mathbf{p}_i(\mathbf{w}_k; \mathbf{x}_i), y_i).$$

After that, from the Assumption 4.1, we have that

$$\begin{aligned} F(\mathbf{w}_{k+1}) &\leq F(\mathbf{w}_k) + \langle \nabla F(\mathbf{w}_k), \mathbf{w}_{k+1} - \mathbf{w}_k \rangle \\ &\quad + \frac{L}{2} \|\mathbf{w}_{k+1} - \mathbf{w}_k\|_2^2. \end{aligned}$$

Next, we investigate the product $\langle \nabla F(\mathbf{w}_k), \mathbf{w}_{k+1} - \mathbf{w}_k \rangle$. Before going into the details, in the subsequent part of this section, we make the following conventions: $\mathcal{H}(n_i) \triangleq \mathcal{H}(n_i; \mathcal{A}(n_i); \tau)$, $\text{CE}_k(i) \triangleq \text{CE}(\mathbf{p}_i(\mathbf{w}_k; \mathbf{x}_i), y_i)$ and $\alpha(i) \triangleq \alpha(\mathbf{x}_i, \hat{y}_i, y_i)$. With all these abbreviations, according to the approximation assumption of the semantic homogeneity $\mathcal{H}(n_i; \mathcal{A}(n_i); \tau)$, i.e., $|\mathcal{H}(n_i; \mathcal{A}(n_i); \tau) - \beta \cdot \alpha(\mathbf{x}_i, \hat{y}_i, y_i)| \leq \epsilon$, we can show that

$$\begin{aligned} &\mathbb{E}(\langle \nabla F(\mathbf{w}_k), \mathbf{w}_{k+1} - \mathbf{w}_k \rangle) \\ &= -\mathbb{E} \left(\left\langle \nabla F(\mathbf{w}_k), \frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathcal{H}(n_i) \nabla \text{CE}_k(i) \right\rangle \right) \\ &= -\beta \mathbb{E} \left(\left\langle \nabla F(\mathbf{w}_k), \mathbb{E} \left(\frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \alpha(i) \nabla \text{CE}_k(i) \mid \mathbf{w}_k \right) \right\rangle \right) \\ &\quad + \mathbb{E} \left(\left\langle \nabla F(\mathbf{w}_k), \frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\beta \alpha(i) - \mathcal{H}(n_i)) \nabla \text{CE}_k(i) \right\rangle \right) \\ &= -\beta \eta \cdot \mathbb{E}(\|\nabla F(\mathbf{w}_k)\|_2^2) \\ &\quad + \mathbb{E} \left(\left\langle \nabla F(\mathbf{w}_k), \frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\beta \alpha(i) - \mathcal{H}(n_i)) \nabla \text{CE}_k(i) \right\rangle \right) \\ &\leq -\beta \eta \cdot \mathbb{E}(\|\nabla F(\mathbf{w}_k)\|_2^2) + \eta \epsilon G^2 \end{aligned}$$

where the final equality follows from Eq.(9) and the final inequality comes from $|\mathcal{H}(n_i) - \beta \alpha(i)| \leq \epsilon$, $\|\nabla F(\mathbf{w}_k)\|_2 \leq G$ as well as $\|\nabla \text{CE}_k(i)\|_2 \leq G$.

Subsequently, we investigate $\|\mathbf{w}_{k+1} - \mathbf{w}_k\|_2^2$, that is,

$$\begin{aligned} \mathbb{E}(\|\mathbf{w}_{k+1} - \mathbf{w}_k\|_2^2) &= \mathbb{E} \left(\left\| \frac{\eta}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathcal{H}(n_i) \nabla \text{CE}_k(i) \right\|_2^2 \right) \\ &\leq 3\eta^2 \mathbb{E} \left(\left\| \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\mathcal{H}(n_i) - \beta \alpha(i)) \nabla \text{CE}_k(i) \right\|_2^2 \right) \\ &\quad + 3\eta^2 \mathbb{E} \left(\left\| \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \beta \alpha(i) \nabla \text{CE}_k(i) - \beta \nabla F(\mathbf{w}_k) \right\|_2^2 \right) \\ &\quad + 3\eta^2 \mathbb{E}(\|\beta \nabla F(\mathbf{w}_k)\|_2^2) \end{aligned}$$

Due to $|\mathcal{H}(n_i) - \beta \cdot \alpha(i)| \leq \epsilon$, we thus have that

$$\mathbb{E} \left(\left\| \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\mathcal{H}(n_i) - \beta \alpha(i)) \nabla \text{CE}_k(i) \right\|_2^2 \right) \leq \epsilon^2 G^2.$$

From Eq.(9) and each input-label pair $\xi_i \triangleq (\mathbf{x}_i, y_i)$ is independently drawn from an underlying distribution $\mathcal{Q}$, we also have that

$$\mathbb{E} \left(\left\| \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \beta \alpha(i) \nabla \text{CE}_k(i) - \beta \nabla F(\mathbf{w}_k) \right\|_2^2 \right) \leq \beta^2 \frac{G^2}{|\mathcal{S}|},$$

Table 4. Overview of the datasets used in this study.

Dataset	# Nodes	# Edges	# Feat.	# Classes	Homophily	Avg. degree
Cora	2708	5278	1433	7	0.81	3.90
CiteSeer	3327	4552	3703	6	0.74	2.74
PubMed	19 717	44 324	500	3	0.80	4.50
A-Photo	7650	238 162	745	8	0.83	31.13
DBLP	17 716	105 734	1639	4	0.83	5.97
Flickr	7575	239 738	12 047	9	0.24	63.30

where the final inequality from the truth that $\text{Var}(\frac{\sum_{i=1}^n X_i}{n}) = \frac{\text{Var}(X_i)}{n}$ for a sequence of i.i.d. random variables $X_1, \dots, X_n$.

As a result, we have that

$$\begin{aligned}
& \mathbb{E}(F(\mathbf{w}_{k+1}) - F(\mathbf{w}_k)) \\
& \leq \mathbb{E}(\langle \nabla F(\mathbf{w}_k), \mathbf{w}_{k+1} - \mathbf{w}_k \rangle) + \frac{L}{2} \mathbb{E}(\|\mathbf{w}_{k+1} - \mathbf{w}_k\|_2^2) \\
& \leq \left(\frac{3L}{2} \eta^2 \beta^2 - \eta \beta \right) \mathbb{E}(\|\nabla F(\mathbf{w}_k)\|_2^2) \\
& \quad + \eta G^2 \epsilon + \frac{3L \eta^2 G^2}{2} \left(\epsilon^2 + \frac{\beta^2}{|S|} \right) \\
& \leq -\frac{\mu}{3L} \mathbb{E}(F(\mathbf{w}_k) - F(\mathbf{w}^*)) + \frac{G^2}{3\beta L} \epsilon + \frac{G^2}{6L} (\epsilon^2 + \beta^2 \epsilon),
\end{aligned}$$

where the final inequality follows from $\eta \triangleq \frac{1}{3\beta L}$, $|S| \triangleq \mathcal{O}(\frac{1}{\epsilon})$ and Assumption 4.2.

Therefore, we have that

$$\begin{aligned}
& \mathbb{E}(F(\mathbf{w}_{k+1}) - F(\mathbf{w}^*)) \leq \left(1 - \frac{\mu}{3L}\right) \mathbb{E}(F(\mathbf{w}_k) - F(\mathbf{w}^*)) \\
& \quad + \frac{G^2}{3\beta L} \epsilon + \frac{G^2}{6L} (\epsilon^2 + \beta^2 \epsilon) \\
& \leq \dots \\
& \leq \left(1 - \frac{\mu}{3L}\right)^k \mathbb{E}(F(\mathbf{w}_1) - F(\mathbf{w}^*)) + \frac{G^2}{\beta \mu} \epsilon \\
& \quad + \frac{G^2}{2\mu} (\epsilon^2 + \beta^2 \epsilon) \\
& = \mathcal{O}\left(\left(1 - \frac{\mu}{3L}\right)^k + \frac{\beta^2 G^2}{\mu} \epsilon\right).
\end{aligned}$$

B. Details about the Datasets

Our experimental evaluation is conducted on six benchmark datasets, which are commonly used in the field of graph machine learning. An overview of their statistical properties is provided in Table 4.

Citation Networks. We utilize three widely recognized citation network datasets:

- Cora (McCallum et al., 2000) is a citation network dataset comprised of scientific publications from the machine learning field. Papers are represented as nodes and citations as edges. The node features are binary vectors indicating the presence of words from a predefined dictionary. Each paper is categorized into one of seven research areas.
- CiteSeer (Giles et al., 1998) is another widely-used citation network where nodes represent scientific publications and edges denote citations. Node features are binary word vectors. The publications are classified into six distinct research categories.
- PubMed (Sen et al., 2008) is a larger citation network consisting of articles from the PubMed database related to

diabetes. Nodes are publications and edges are citations. Node features are TF-IDF weighted word vectors from a specific dictionary. The papers are classified into one of three classes.

Co-purchase and Co-authorship Networks. We also include two datasets based on co-occurrence:

- A-Photo (Shchur et al., 2018) is a segment of the Amazon co-purchase network where nodes represent products and edges indicate that two products are frequently bought together. Node features are derived from product reviews, and the labels correspond to product categories.
- DBLP (Pan et al., 2016) is a co-authorship network from the computer science bibliography. Nodes represent authors, and an edge connects two authors if they have co-authored at least one publication. The node features are based on the keywords of the authors’ papers, and the labels indicate the author’s primary field of research.

Social Network. Finally, to evaluate performance on a different graph structure, we use a social network dataset:

- Flickr (Yang et al., 2020) is a social network where nodes represent users and edges indicate a friendship or contact relationship between them. Node features are created from user-specified tags, and the labels represent the user’s interest groups. This dataset is known for its lower homophily ratio compared to the other datasets.

C. Details about the Baselines

We compare our proposed method against a comprehensive suite of baselines. These methods are grouped into three categories: a vanilla GNN model, traditional Learning with Label Noise (LLN) methods adapted for graphs, and methods specifically designed for Graph Neural Networks under Label Noise (GLN).

Vanilla GNN. This serves as the fundamental baseline to evaluate the performance degradation caused by label noise.

- GCN (Kipf & Welling, 2016) is a standard Graph Convolutional Network that aggregates information from neighboring nodes using a simplified spectral filtering approach. We use it as our foundational backbone and primary baseline.

Learning with Label Noise (LLN) Methods. These are representative methods from the broader field of learning with noisy labels, which we adapt to the graph domain to assess their applicability.

- **S-model** (Goldberger & Ben-Reuven, 2017) adds a noise adaptation layer to the network output, which models the noise transition matrix and learns it jointly with the classifier parameters.
- **Coteaching** (Han et al., 2018) maintains two identical networks. In each training batch, each network selects samples with small losses and feeds these presumably clean samples to its peer for updating, thus preventing error accumulation.
- **JoCoR** (Wei et al., 2020) also trains two networks but uses a joint loss function. This loss includes the individual supervised losses and a consistency regularization term that encourages the two networks to produce similar predictions.
- **Forward** (Patrini et al., 2017) is a loss correction method that multiplies the network’s softmax output by an estimated noise transition matrix before computing the loss, effectively correcting the predictions.
- **Backward** (Patrini et al., 2017) is an alternative loss correction approach that adjusts the gradients during backpropagation based on the estimated noise transition matrix.
- **SCE** (Wang et al., 2019) proposes a noise-robust loss function, Symmetric Cross Entropy, which combines the standard cross-entropy with a reverse cross-entropy term, making it less susceptible to noisy labels.

Graph Neural Networks under Label Noise (GLN) Methods. This category includes state-of-the-art methods specifically developed to handle noisy labels in graph-structured data.

- **NRGNN** (Dai et al., 2021) refines the graph structure by adding edges between nodes with similar features and then generates pseudo-labels on this refined graph to provide additional, more reliable supervision signals.
- **RTGNN** (Qian et al., 2023) extends NRGNN by using a small-loss criterion to differentiate between trustworthy and untrustworthy nodes, then selectively corrects the labels of the latter to mitigate noise propagation.
- **CP** (Zhang et al., 2020) defends against adversarial label-flipping by leveraging graph community information. It uses community-level consistency as a robust signal to guide node classification.
- **CLNode** (Wei et al., 2023) introduces a curriculum learning strategy that measures the quality of training nodes from multiple perspectives and schedules the training process from easy (clean) to hard (noisy) samples.
- **PIGNN** (Du et al., 2021) utilizes noise-robust pairwise interaction (PI) labels. It estimates whether pairs of nodes share the same true label based on embedding similarity and uses these PI labels to regularize the training of the node classifier.
- **UnionNET** (Li et al., 2021) devises a reweighting scheme where the importance of a labeled node is determined by its label consistency with its local neighborhood, aggregated through random walks and an attention mechanism.
- **CR-GNN** (Li et al., 2024c) leverages contrastive learning on augmented graph views (generated via feature masking and edge dropping) to learn robust representations that are invariant to local structural changes and less affected by label noise.

D. Additional Details of Implementation

Training Objective. As mentioned in the main text, to regularize representations against label noise, we incorporate an auxiliary contrastive task. The full objective function is:

$$\mathcal{L}_{total} = \mathcal{L} + \lambda \mathcal{L}_{CL}, \quad (10)$$

where $\mathcal{L}$ is the weighted cross-entropy loss defined in Eq.(7), and $\mathcal{L}_{CL}$ is the contrastive loss.

Contrastive Learning details. We adopt the GraphCL (You et al., 2020) framework. For each iteration, we generate two augmented views of the graph by stochastic *edge dropping* with a probability p . Let $\mathbf{z}_i^{(1)}$ and $\mathbf{z}_i^{(2)}$ be the latent representations of node v_i from the two views. The contrastive loss $\mathcal{L}_{CL}$ is formulated using the InfoNCE objective:

$$\ell_i = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)})/\tau_{cl})}{\sum_{j=1, j \neq i}^N \exp(\text{sim}(\mathbf{z}_i^{(1)}, \mathbf{z}_j^{(2)})/\tau_{cl})}, \quad \mathcal{L}_{CL} = \frac{1}{N} \sum_{i=1}^N \ell_i, \quad (11)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity.

E. Additional Experimental Results

E.1. Additional Results under Different Noise Levels

Performance under Additional Noise Types on Cora and CiteSeer. To complement the results under uniform noise presented in the main paper, Figure 7 illustrates the performance of DREAM and competing baselines on the Cora and CiteSeer datasets, specifically under Pair and Asymmetric noise with rates varying from 0 to 0.5.

As can be observed from the figure, our proposed DREAM method consistently outperforms all compared baselines across both datasets and both noise types. This superiority is particularly evident as the noise rate increases. While the performance of baseline methods like GCN, NRGNN, and PIGNN degrades significantly at higher noise levels, DREAM demonstrates much greater robustness. It maintains a stable and high accuracy, indicating that our relation-informed dynamic optimization framework can effectively identify and mitigate the impact of noisy labels regardless of the noise pattern. For instance, in Figure 7d, under 50% asymmetric noise on the CiteSeer dataset, DREAM’s accuracy is substantially higher than the best-performing baseline, showcasing its effectiveness in challenging noisy scenarios. These comprehensive results further validate the robustness and generalizability of the proposed DREAM method.

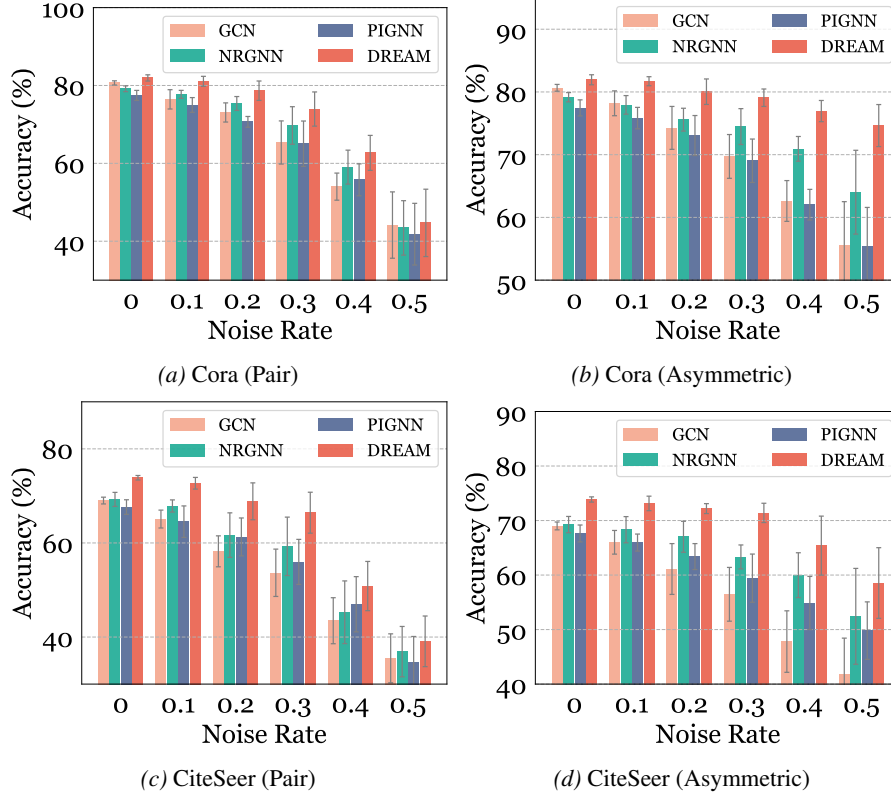


Figure 7. Performance under different levels of label noise. (a) Pair noise on Cora. (b) Asymmetric noise on Cora. (c) Pair noise on CiteSeer. (d) Asymmetric noise on CiteSeer.

Performance on Diverse Graph Structures. To further demonstrate the robustness and generalizability of our method, Figure 8 presents the comprehensive performance comparison on the PubMed, A-Photo, and Flickr datasets. These datasets represent a wider variety of graph structures, including larger scale (PubMed), higher density (A-Photo), and significant heterophily (Flickr).

The results are remarkably consistent with our previous findings. Across all nine experimental settings shown in the figure, DREAM consistently achieves superior performance over the baseline methods. As the noise rate increases from 0 to 0.5, the performance of all methods declines, but the accuracy of DREAM degrades much more gracefully. This trend is especially pronounced on the challenging Flickr dataset (Figure 8(g-i)), which is known for its low homophily. On such heterophilic graphs, where node labels are often dissimilar from their neighbors, many GNN-based methods struggle. However, DREAM maintains a substantial performance margin, validating that its dual-standard anchor selection mechanism can effectively capture reliable relational information beyond simple homophily assumptions. Collectively, these extensive results strongly support the conclusion that DREAM is a robust and widely applicable solution for learning on graphs with label noise.

Performance under Extreme Noise Levels. To answer concerns regarding highly corrupted observed labels, we further push the noise rate to extreme regimes (50%–70%) on the Cora dataset under uniform noise. As demonstrated in Table 5, while standard GNNs and baseline robust methods suffer severe performance degradation (e.g., GCN drops to 32.78% at 70% noise), DREAM maintains a remarkable and consistent advantage, achieving 60.22% accuracy even when 70% of the labels are corrupted. This is because our topology-aware anchors $\mathcal{A}_T$ are label-free, keeping the anchor set informative and mitigating confirmation bias under high corruption.

E.2. Additional Baselines and Heterophilic Graphs

We further evaluate DREAM against ProCon (IJCAI 2025), a recently proposed robust graph neural network against label noise. As shown in Table 6, DREAM consistently outperforms the recent baseline under 30% uniform noise.

Additionally, to thoroughly validate the robustness of DREAM on graphs with lower homophily, we supplement experiments

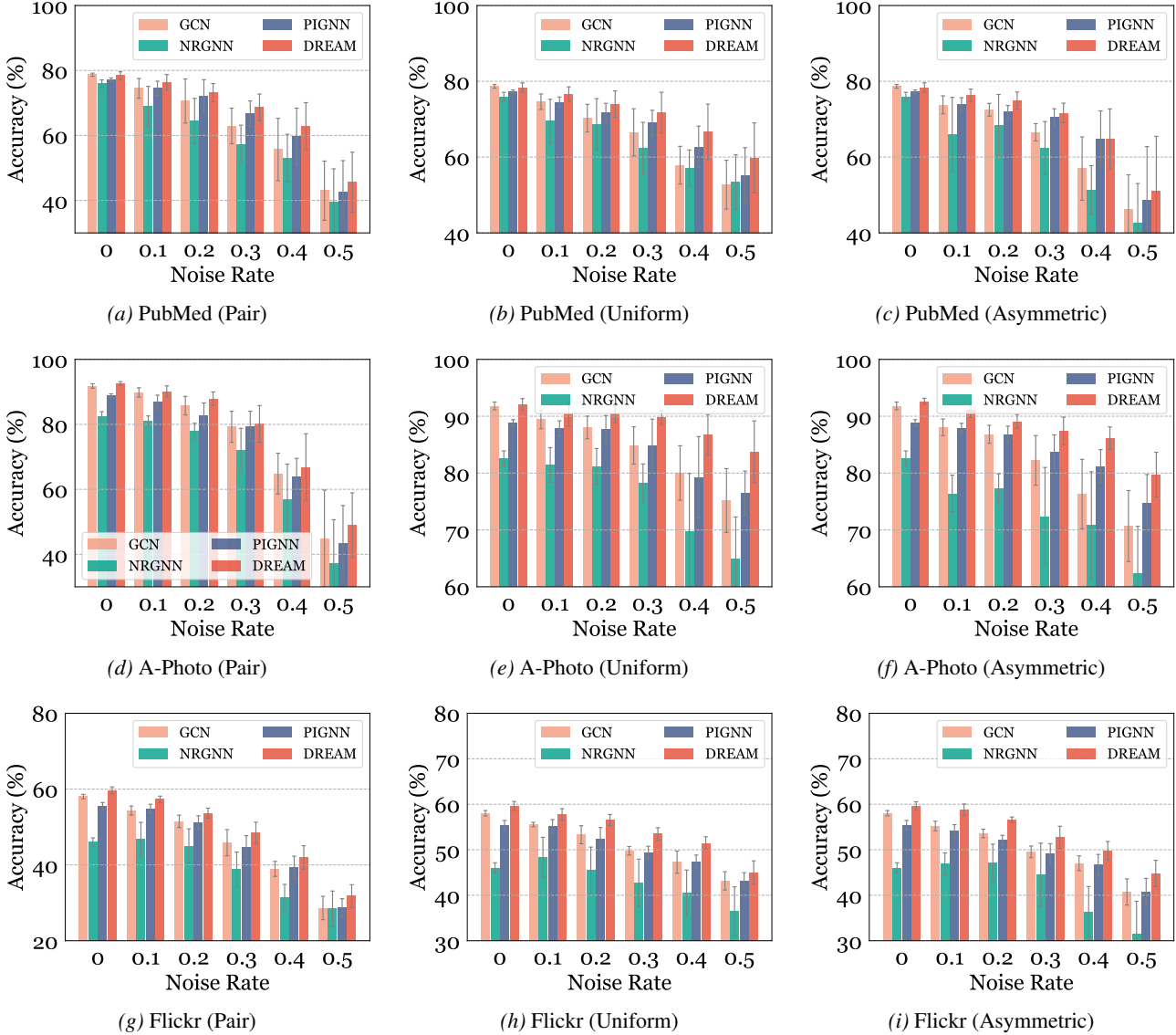


Figure 8. Performance comparison under different levels and types of label noise on PubMed, A-Photo, and Flickr. Rows (top to bottom) correspond to datasets: PubMed, A-Photo, Flickr. Columns (left to right) correspond to noise types: Pair, Uniform, and Asymmetric.

on the Chameleon ($h = 0.24$) and Squirrel ($h = 0.22$) datasets under 30% uniform noise. The results in Table 7 show that DREAM outperforms state-of-the-art baselines like NRGNN and PIGNN. Topology-aware anchors ($\mathcal{A}_T$) are label-free and rely strictly on geodesic distance, which successfully mitigates the impact of low homophily.

E.3. Performance under Sparse Labels and Deeper Architectures

We test the compatibility of DREAM in sparse-label regimes. Table 8 reports the performance on Cora under 30% noise with 5, 10, and 20 (default) labels per class. DREAM (Full) maintains a strong lead over basic GCN and its contrastive learning variant. The auxiliary contrastive learning provides substantial complementary benefits when label availability is highly restricted.

We also verify that DREAM generalizes consistently to deeper network architectures. Table 9 shows performance using 3-layer and 4-layer GCNs under 30% uniform noise. DREAM's reweighting operates on node representations independently of encoder depth, granting robust improvements regardless of network layers.

Table 5. Classification accuracy under extreme uniform noise rates (50%–70%) on Cora.

Noise Rate	50%	60%	70%
GCN	54.42	39.86	32.78
NRGNN	64.90	51.37	45.36
PIGNN	55.28	41.39	34.30
DREAM (Ours)	75.63	66.20	60.22

Table 6. Comparison with recent baseline ProCon under 30% uniform noise.

Dataset	ProCon	DREAM (Ours)
Cora	78.86	80.40
CiteSeer	68.11	71.06

E.4. Extended Ablation Studies on Contrastive Learning

To detail the contribution of the contrastive learning component, we compare DREAM (Full) with its variant without contrastive learning (V6: w/o CL) and a model trained solely with contrastive learning (Only CL). As shown in Table 10, the dynamic reweighting strategy remains the primary contributor to performance improvement, while contrastive learning provides complementary regularization.

E.5. Further Hyperparameter and Design Analysis

Sensitivity to τ and d_{max} . In Table 11 and Table 12, we report the performance sensitivity of DREAM concerning the temperature scaling parameter τ and the maximum geodesic distance d_{max} for candidate set construction, evaluated on Cora and CiteSeer under 30% uniform noise. Performance remains stable across a wide range of values. The choice of $d_{max} = 4$ aligns appropriately with the 2-hop neighbor receptive field of the default 2-layer GCN.

Table 11. Sensitivity analysis of the temperature parameter τ .

τ	0.01	0.02	0.04 (Def.)	0.08	0.16
Cora	66.25	66.37	80.40	73.32	72.42
CiteSeer	63.07	64.54	71.06	62.73	59.63

Adaptive vs. Fixed Anchor Selection. We explored adaptively adjusting k_P/k_T per node based on local density (i.e., $k_P = \min(15, \lfloor |\mathcal{C}_P|/3 \rfloor)$, $k_T = \min(10, \lfloor |\mathcal{C}_T|/3 \rfloor)$). As seen in Table 13, the adaptive configuration performs slightly below the fixed setting, indicating that the default fixed selection bounds are sufficiently robust.

Class Purity of Topology-Aware Anchors. We analyzed the class purity of topology-aware anchors ($\mathcal{A}_T$) on a homophilic graph (Cora) versus a heterophilic graph (Flickr). Purity is measured by the fraction of anchors sharing the true underlying class label. Table 14 shows that topology anchors carry strong label-level information on homophilic graphs but remain useful via representation-level signals on heterophilic graphs, further validating the necessity of a dual-standard framework.

E.6. Additional Results on the Evolution of Semantic Homogeneity Scores

In this section, we provide additional results on the evolution of the semantic homogeneity scores under 30% asymmetric label noise. The results comparing correctly labeled nodes (clean) and mislabeled nodes (noisy) are shown in Figure 9. The results under asymmetric label noise exhibit similar patterns to the uniform label noise in Section 5.5. Initially, the two groups of nodes (clean and noisy) have similar semantic homogeneity scores. As the training proceeds, clean nodes tend to have higher homogeneity scores and noisy nodes tend to have lower scores. This shows that the proposed DREAM is able to separate correctly labeled nodes (clean) and mislabeled nodes (noisy).

Furthermore, we quantitatively track the average $\mathcal{H}$ -scores across varying noise severities at convergence on Cora (Table 15). The $\sim 3\times$ clean-vs-noisy gap persists consistently even at a 60% extreme noise rate, empirically validating that the learned

Table 7. Performance on heterophilic graphs under 30% uniform noise.

Method	DREAM	NRGNN	PIGNN
Actor ($h = 0.22$)	29.74	27.97	26.11
Chameleon ($h = 0.24$)	63.67	49.54	58.35
Squirrel ($h = 0.22$)	46.91	34.09	44.80

Table 8. Performance under sparse label settings on Cora (30% uniform noise).

Labels/class	5	10	20 (Def.)
GCN	52.85	59.43	71.06
GCN + CL	48.68	60.97	71.27
DREAM w/o CL	51.82	63.30	73.95
DREAM (Full)	63.72	75.95	80.40

reliability score distinguishes clean from noisy elements independent of the absolute noise corruption ratio.

Table 15. Average $\mathcal{H}$ -scores at convergence under extreme noise rates on Cora.

Noise Rate	30%	40%	50%	60%
Clean Nodes	0.049	0.047	0.038	0.039
Noisy Nodes	0.0072	0.013	0.0081	0.0126

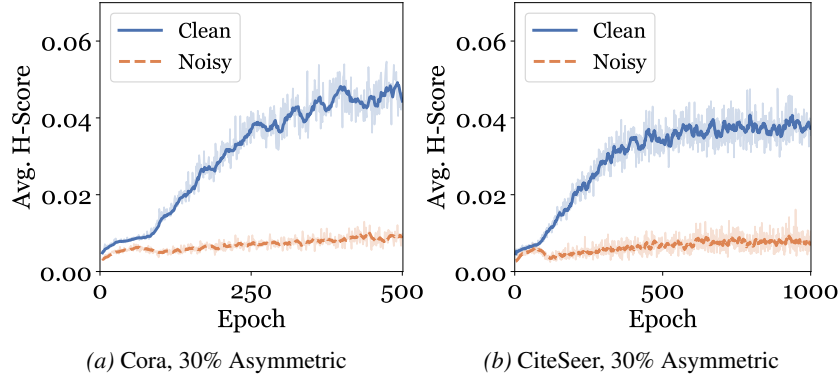


Figure 9. Evolution of semantic homogeneity scores of clean and noisy nodes on Cora and CiteSeer under 30% asymmetric label noise. DREAM successfully separates clean from noisy nodes.

E.7. Additional Results on Visualization of Learned Representations

In this section, we provide additional results on the visualization of learned representations of the proposed DREAM in comparison to the GCN baseline. The results are shown in Figure 10, where we perform experiments on the CiteSeer dataset under 40% uniform label noise and use t-SNE (Maaten & Hinton, 2008) for visualization. The results are similar to the experiments in Section 5.5, and we can see that the proposed DREAM yields more condensed and clustered representations in the hidden space compared to the GCN baseline.

Table 9. Generalization to deeper architectures (30% uniform noise).

Architecture	Dataset	GCN	DREAM
3-layer GCN	Cora	72.52	77.23
	CiteSeer	55.74	68.58
4-layer GCN	Cora	72.08	73.99
	CiteSeer	54.97	65.79

Table 10. Ablation (V6) on contrastive learning (30% uniform noise).

Dataset	DREAM (Full)	V6: w/o CL	Only CL
Cora	80.40	73.95	71.27
CiteSeer	71.06	64.43	57.07

Table 12. Sensitivity analysis of maximum geodesic distance d_{max} .

d_{max}	2	3	4 (Def.)	5	6
Cora	79.90	80.18	80.40	78.99	78.94
CiteSeer	69.64	69.97	71.06	70.70	70.39

Table 13. Fixed vs. Adaptive Anchor Selection (k_P, k_T).

Dataset	Fixed k	Adaptive k
Cora	80.40	79.91
CiteSeer	71.06	70.53

Table 14. Class purity of topology-aware anchors ($\mathcal{A}_T$).

Dataset	Cora (Homophilic)	Flickr (Heterophilic)
Purity of $\mathcal{A}_T$	0.3788	0.1112
Random Baseline	0.1796	0.1113

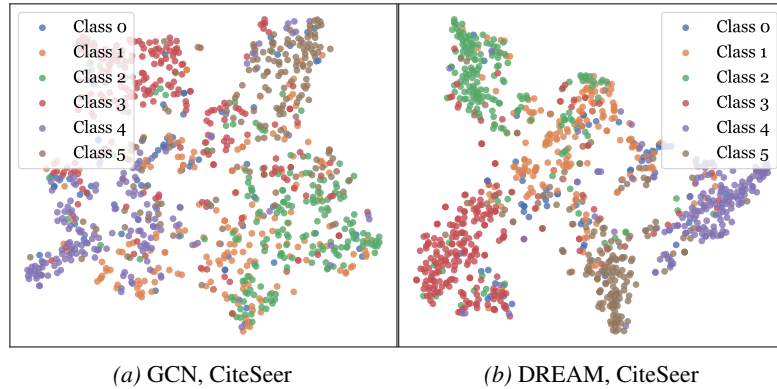


Figure 10. Learned representations under 40% uniform label noise on CiteSeer. The proposed DREAM yields more condensed clusters and better separates different classes compared to GCN.